

Yichen Hu

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EDUCATION

- **University of California, Davis** Davis, CA, USA
Ph.D. Candidate in Statistics Sep 2023 – Present
Advisor: Dr. Xiukai Ding.
Research interests: high-dimensional statistics, random matrix theory, manifold learning, graph networks, and applications to data science and machine learning.
- **University of California, Davis** Davis, CA, USA
Master of Science in Statistics Sep 2021 – Jun 2023
- **Shanghai University of Finance and Economics** Shanghai, China
Bachelor of Science in Statistics Sep 2016 – Jul 2020

RESEARCH

- Xiukai Ding, **Yichen Hu**, Zhenggang Wang (2025). Two Sample Test for Covariance Matrices in Ultra-High Dimension. *Journal of the American Statistical Association*.
We develop an adaptive testing procedure for equality of covariance matrices in ultra-high-dimensional settings, combining sample splitting with eigenvalue-based statistics under weak structural assumptions.
An accompanying R package, `UHDtst`, implements the proposed methodology.
- Xiukai Ding, **Yichen Hu** (2025). On the Edge Eigenvalues of Sparse Random Geometric Graphs. Submitted to *Annals of Applied Probability*.
We study the edge eigenvalues of sparse random geometric graphs generated by multivariate Gaussian samples, and introduce a smoothing-matching approach that links graph eigenvalues to those of a continuum differential operator.
- **Yichen Hu**, Xiukai Ding, Jian Kang. Adaptive Sparsity-Aware Projection-Assisted Spectral Embedding for High-Dimensional Noisy Data with Nonlinear Structures. *In progress*.
We propose an adaptive projection-assisted spectral embedding method for high-dimensional noisy data with nonlinear structure and latent sparsity, aiming to denoise pairwise geometry and recover meaningful low-dimensional embeddings across different signal-to-noise regimes.

AWARDS

- IMS Hannan Graduate Student Travel Award, 2026
- Outstanding Graduate Student Teaching Award, Department of Statistics, University of California, Davis, 2024–25
- Outstanding Graduate Student Teaching Award, Department of Statistics, University of California, Davis, 2023–24
- Outstanding Master's Student Award, Department of Statistics, University of California, Davis, 2021–22